MUSC: ZERO-SHOT INDUSTRIAL ANOMALY CLASSIFICATION AND SEGMENTATION WITH MUTUAL SCORING OF THE UNLABELED IMAGES

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ABSTRACT

This paper studies zero-shot anomaly classification (AC) and segmentation (AS) in industrial vision. We reveal that the abundant normal and abnormal cues implicit in unlabeled test images can be exploited for anomaly determination, which is ignored by prior methods. Our key observation is that for the industrial product images, the normal image patches could find a relatively large number of similar patches in other unlabeled images, while the abnormal ones only have a few similar patches. We leverage such a discriminative characteristic to design a novel zero-shot AC/AS method by Mutual Scoring (MuSc) of the unlabeled images, which does not need any training or prompts. Specifically, we perform Local Neighborhood Aggregation with Multiple Degrees (LNAMD) to obtain the patch features that are capable of representing anomalies in varying sizes. Then we propose the Mutual Scoring Mechanism (MSM) to leverage the unlabeled test images to assign the anomaly score to each other. Furthermore, we present an optimization approach named Re-scoring with Constrained Image-level Neighborhood (RsCIN) for image-level anomaly classification to suppress the false positives caused by noises in normal images. The superior performance on the challenging MVTec AD and VisA datasets demonstrates the effectiveness of our approach. Compared with the state-of-the-art zero-shot approaches, MuSc achieves a 21.1% PRO absolute gain (from 72.7% to 93.8%) on MVTec AD, a 19.4% pixel-AP gain and a 14.7% pixel-AUROC gain on VisA. In addition, our zeroshot approach outperforms most of the few-shot approaches and is comparable to some one-class methods. Code is available at https://github.com/xrli-U/MuSc.

1 INTRODUCTION

Industrial anomaly classification (AC) and anomaly segmentation (AS) are fundamental tasks in computer vision with diverse applications. Real-world anomalies may appear on a wide range of objects and textures. The high diversity of anomalies makes the AC/AS task challenging.

Many excellent unsupervised AC/AS approaches, e.g., (Zavrtanik et al., 2022; Zhang et al., 2023; Zavrtanik et al., 2021; Liu et al., 2023), have been proposed. They typically rely on the whole normal dataset for training, thus called as full-shot methods. In contrast, RegAD (Huang et al., 2022) and GraphCore (Xie et al., 2023) leverage a few normal images only, known as few-shot AC/AS setting, and also achieve promising accuracy. All the methods above can be categorized as one-class AC/AS approaches, and share a similar pipeline as shown in Fig. 1(a). Recently, WinCLIP (Jeong et al., 2023) and APRIL-GAN (Chen et al., 2023) propose zero-shot AC/AS, which breaks new ground by using text prompts for anomaly measurement, as shown in Fig. 1(b).

In these existing approaches, the test images are compared to the additionally introduced information, such as normal training images or text prompts. However, a rich amount of normal information

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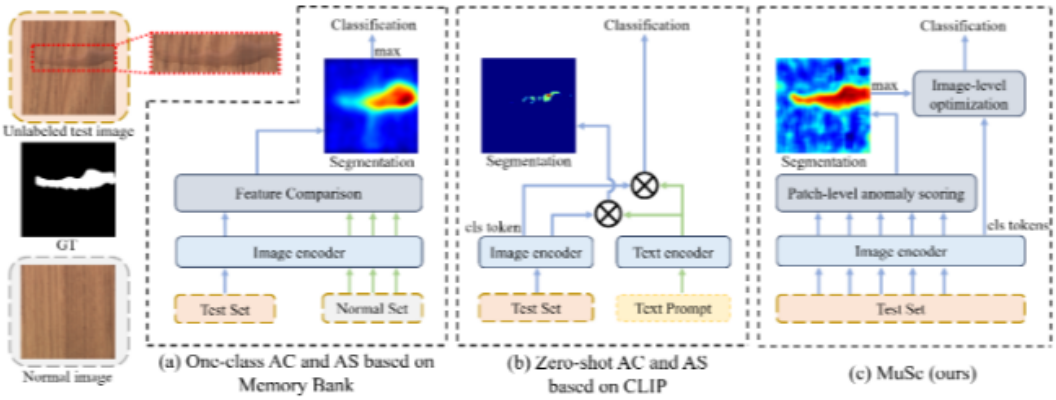


Figure 1: (a) One-class AC and AS based on a memory bank, which requires many normal reference images. (b) Zero-shot AC and AS based on CLIP, which relies on additional text prompts. (c) Our MuSc only leverages the unlabeled test images from patch-level and image-level.

implicit in the unlabeled test images is rarely used. According to statistics, the normal pixels accounted for 97.26% in the test images of MVTec AD dataset and 99.45% in VisA dataset. Besides, due to the randomness and unpredictability of the anomalies, the abnormal regions may be dissimilar from each other, even though they belong to the same anomaly type. Therefore, both normal and abnormal cues in unlabeled test images can be exploited for anomaly determination.

Based on these observations, we reveal that the anomaly can be detected by comparing it with other unlabeled test images, and hence we design a novel zero-shot AC/AS approach, named MuSc, which does not need any normal training images or prompts. As shown in Fig. 1(c), our approach consists of a patch-level representation extractor, a patch-level anomaly scoring approach, and an image-level optimization. First, we perform Local Neighborhood Aggregation with Multiple Degrees (LNAMD) on the patch token to express each image patch, which is capable of modeling anomalies in varying sizes and is suitable for AC and AS tasks. Next, we introduce a Mutual Scoring Mechanism (MSM), which leverages the unlabeled test images to assign the anomaly scores to each other. Such a simple mechanism generates a high-quality anomaly score for each image region. It can be credited to the observation that the normal image patches could find a large number of similar patches in other unlabeled images while the abnormal ones only have a few similar patches. Furthermore, we explore the relationship between image-level features and propose the Re-scoring with Constrained Image-level Neighborhood (RsCIN) to optimize the anomaly classification. Extensive experimental evaluations on MVTec AD and VisA datasets demonstrate the effectiveness of our approach. Compared with the state-of-the-art zero-shot AC/AS approaches, we achieve +6.0% AUROC classification metrics on MVTec AD and +14.7% increase on VisA. For anomaly segmentation, we obtain +21.1% PRO and +21.9% AP gains on MVTec AD and +19.4% AP gains on VisA. We further emphasize that our zero-shot approach outperforms most existing few-shot methods in 4-shot case and is comparable to several full-shot methods.

In summary, the major contributions of our work are:

- To the best of our knowledge, we propose the first method that only uses the unlabeled test images for industrial anomaly classification and segmentation.
- We reveal the potential capability of normal and abnormal patches contained in unlabeled images. It inspires us to propose a brand new AC/AS method based on mutual scoring of unlabeled images, which is a new mechanism for the community.
- Our approach significantly surpasses the existing zero-shot AC and AS approaches. In addition, our approach outperforms most of the few-shot approaches and is comparable to some full-shot methods.

2 RELATED WORKS

Vision transformer. Vision transformer (ViT) (Dosovitskiy et al., 2020) is widely used in multi-modal models and large vision models such as CLIP (Radford et al., 2021) and DINO (Caron et al., 2021). These models produce high-quality patch-level and image-level features. We find that the patch-level features have the limitation of detecting industrial anomalies of varying sizes. Swin transformer (Liu et al., 2021) proposes the varied-size window attention to calculate attention inside local windows of varying sizes, which can detect diverse size objects. We propose a simple method LNAMD to optimize the patch-level features so that it can detect varying size anomalies.

Industrial anomaly classification and segmentation. In the industrial vision field, previous works explore one-class AC/AS, such as PatchCore (Roth et al., 2022), PaDiM (Defard et al., 2021) and AST (Rudolph et al., 2023), which require a lot of normal reference images. Recently, some zero-shot methods have worked well. Most of them rely on text prompts for the image-text alignment such as WinCLIP (Jeong et al., 2023), Variance Vigilance Vanguard (Baugh et al., 2023) and SAA (Cao et al., 2023). In the few-shot setting, RegAD (Huang et al., 2022) and GraphCore (Xie et al., 2023) focus on data augmentation to estimate the normal feature distribution by a few reference normal images. These methods ignore the rich information in the unlabeled images. In the medical image analysis field, some methods like DDAD (Cai et al., 2023) and SRR (Yoon et al., 2021) integrate the unlabeled images into the normal image set for model training, which also needs to select a set of normal images. In the zero-shot AC/AS field, the method (Aota et al., 2023) explores the relationship between the patches inside one test texture image. ACR (Li et al., 2023) proposes a new adaptation strategy without human involvement and outperforms other methods. We propose the first zero-shot AC/AS method that only uses the unlabeled images.

Manifold learning. In high-dimensional manifolds, only local space maintains the nature of Euclidean space, so it is inaccurate to calculate the distance directly in high-dimensional manifolds. The idea of manifold learning methods (Zhou et al., 2003; Wang and Tu, 2012) is to construct a new embedding space where the distance metric matches the manifold structure. We find that the image-level features satisfy the conditions for higher dimensional manifolds, and hence we design the RsCIN based on manifold learning to optimize the pixel-level anomaly classification results.

3 METHOD

As illustrated in Fig. 2, our approach is directly designed on the unlabeled test images. Firstly, we extract the ViT feature for each unlabeled image and perform Local Neighborhood Aggregation with Multiple Degrees (LNAMD) on patch tokens to represent abnormal regions with varying sizes, detailed in Section 3.1. Secondly, we propose a Mutual Scoring Mechanism (MSM) in Section 3.2, in which the unlabeled images assign the anomaly scores to each other, resulting in pixel-level anomaly classification and segmentation. Thirdly, we leverage Re-scoring with Constrained Image-level Neighborhood (RsCIN) to optimize the pixel-level anomaly classification in Section 3.3.

3.1 LOCAL NEIGHBORHOOD AGGREGATION WITH MULTIPLE DEGREES

Given the unlabeled test images $\mathcal{D}_u = \{I_i, i = 1, ..., N\}$. We extract the ViT feature (Dosovitskiy et al., 2020) of I_i. To increase the abnormal modeling ability with varying sizes, we perform local neighborhood aggregation with multiple degrees on the patch tokens of ViT. Specifically, we define the patch tokens as $F_i \in \mathbb{R}^{M \times C}$ of I_i, where M is the patch number, then we reshape F_i to $\sqrt{M} \times \sqrt{M} \times C$. For each patch token, we follow (Roth et al., 2022) to use an adaptive average pooling operation in its $r \times r$ neighborhood to obtain the aggregated patch token $\hat{F}_i^r \in \mathbb{R}^{\sqrt{M} \times \sqrt{M} \times C}$ with different aggregation degree $r \in \{1, 3, 5\}$, shown in the left of Fig. 3, and we reshape it back to $M \times C$. Therefore each aggregated patch token is denoted as $\hat{p}_{i,l}^{m,r} \in \mathbb{R}^{1 \times C}$, where $m \in [1, M]$, we divide the layers of ViT into L stages and $l \in \{1, 2, ..., L\}$ indicates the stage l of ViT. We will integrate multiple aggregation degrees and stages of ViT in the next subsection.

Compared with using one patch size, the patch token with multiple aggregation degrees is suitable for representing anomalies with varying sizes, leading to high-quality anomaly scores even with a simple distance measurement. As the industrial images illustrated in Fig. 3. When $r = 1$, we do not perform

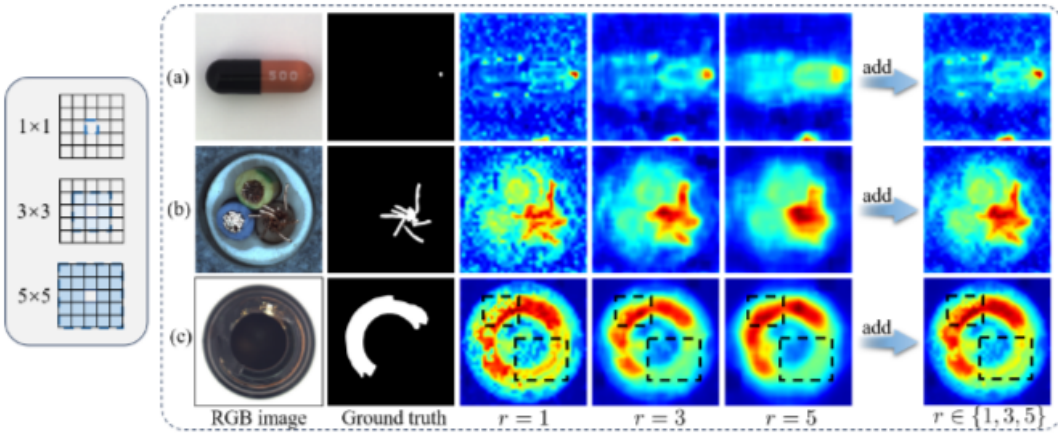


Figure 2: The visualization of anomaly segmentation results with different aggregation degrees r.

LNAMD. The defect has a higher anomaly score in (a), which indicates that smaller r is suitable for detecting small anomalies, while in (c), the large defect has several local false positives and false negatives, indicated by the black dotted box, since the smaller regions cannot fully express the entire defect. When $r = 5$, the false positives and false negatives are solved in (c), demonstrating that the aggregated feature is suited to detect the larger defects, while in (a), the small defect is smoothed and has a lower score. Therefore, using all aggregated patch tokens with different degrees achieves a good balance in detecting the anomaly with a wide range of sizes, which is the common case in real industrial scenes, as shown in the right side of Fig. 3.

3.2 Mutual Scoring Mechanism of the Unlabeled Images

In this subsection, we propose the mutual scoring mechanism to obtain the high-quality patch-level anomaly score by only using the unlabeled images in $\mathcal{D}_u$, which consists of three steps. In the first step, we leverage each image in $\{\mathcal{D}_u \backslash I_i\}$ to assign an anomaly score for each aggregated patch token of a test image I_i in stage l as follows,

$$a_{i,l}^{m,r}(I_j) = \min_n \left\| \hat{p}_{i,l}^{m,r} - \hat{p}_{j,l}^{n,r} \right\|_2 \tag{1}$$

where r is the aggregation degree and (I_j) indicates that image $I_j \in \{\mathcal{D}_u \backslash I_i\}$ is employed for scoring. If the patch token $\hat{p}_{i,l}^{m,r}$ of I_i is similar to any patch token $\hat{p}_{j,l}^{n,r}$ of I_j, the image I_j assigns a small anomaly score to $\hat{p}_{i,l}^{m,r}$. Then we define the scoring vector $A_{i,l}^{m,r} = [a_{i,l}^{m,r}(I_1), a_{i,l}^{m,r}(I_2), ..., a_{i,l}^{m,r}(I_{N-1})]$ of each patch for each l and r according to $\{\mathcal{D}_u \backslash I_i\}$. The statistical histogram of $A_{i,l}^{m,r}$ of all the normal patches in $\mathcal{D}_u$ is given in Fig. 4(a), which indicates that most of the normal patches could find some similar patches in $\mathcal{D}_u$. While (b) gives the histogram of abnormal patches, which shows that most abnormal patches only have a few similar patches in $\mathcal{D}_u$, indicating in the left corner of Fig. 4(b), and they are dissimilar to most of the normal and other abnormal patches in (b).

Based on the above analysis, we claim that most of the unlabeled images in $\mathcal{D}_u$ could assign a higher score for the anomaly patch and a lower score for the normal ones. Hence, directly performing an average operation on $A_{i,l}^{m,r}$ is capable of distinguishing the abnormal patches from the normal patches, shown in Fig. 4(c). For the overlap in (c), we notice that it is caused by some normal patches with appearance variations of different normal images, which are assigned higher scores by the dissimilar patches. To reduce these dissimilar patches in scoring, the second step of MSM is to perform an Interval Average (IA) operation on the minimum $X\%$ of $A_{i,l}^{m,r}$.

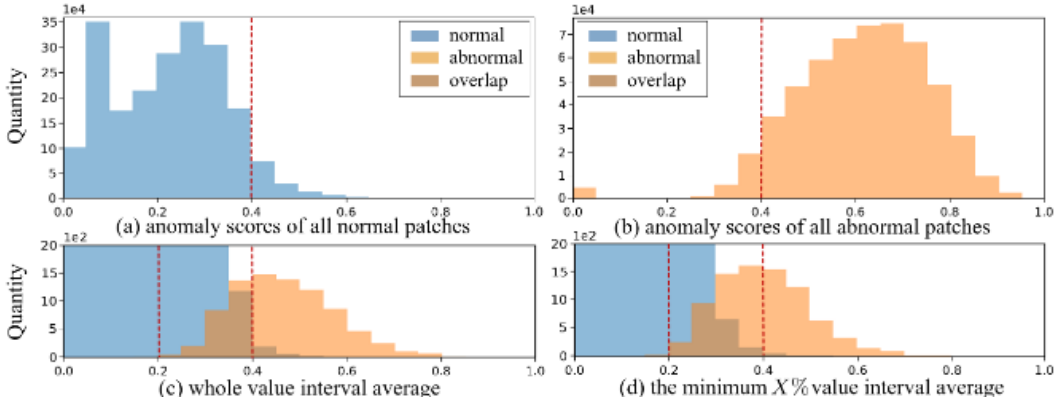


Figure 3: **Top:** The histograms of $A_{i,l}^{m,r}$ of all normal patches (a) and abnormal patches (b). **Bottom:** The histograms of $a_{i,l}^{m,r}$ of normal patches and abnormal patches with the whole value interval (c) and the minimum $X\%$ value interval (d) to average. The blue and orange parts indicate the anomaly scores of normal patches and abnormal patches respectively under $r = 3$ and $l = 2$ setting.

$$\bar{a}_{i,l}^{m,r} = \frac{1}{K} \sum_{k \in [1,K]} a_{i,l}^{m,r}(I_k) \tag{2}$$

where $\bar{\mathcal{I}}$ is the images in the minimum $X\%$ value interval and K is the number of these images. Such a design uses a small number of images to score each image patch, and suppresses the aforementioned issue effectively. As shown in Fig. 4(d), the anomaly score distribution of normal and abnormal patches has a smaller overlap in the score interval of $[0.2, 0.4]$ compared with (c).

Aiming to achieve the fine-grained anomaly score for each image patch, the third step of MSM performs Eq. 1 and Eq. 2 on each combination of stage l and aggregation degree r, respectively, and compute the average of all $\bar{a}_{i,l}^{m,r}$ to achieve the patch-level anomaly score as follows,

$$\mathbf{a}_i^m = \frac{1}{L} \sum_{l \in \{1,...,L\}} \frac{1}{3} \sum_{r \in \{1,3,5\}} \bar{a}_{i,l}^{m,r} \tag{3}$$

The patch-level anomaly score vector of I_i is defined as $\mathcal{A}_i = [\mathbf{a}_i^1, ..., \mathbf{a}_i^M]^\top$. Then we reshape $\mathcal{A}_i \in \mathbb{R}^{M \times 1}$ to $\sqrt{M} \times \sqrt{M} \times 1$ and upsample it to the original resolution of the input image as the anomaly segmentation result. Following existing approaches (Roth et al., 2022), we extract the maximum score $c_i = \max(\mathbf{a}_i^1, ..., \mathbf{a}_i^M)$ of image I_i as its pixel-level anomaly classification score and the AC score vector of the images in $\mathcal{D}_u$ is denoted as $\mathbf{C} = [c_1, ..., c_N]^\top$. See Appendix A.1.1 for more details.

3.3 Classification Re-scoring with Constrained Image-level Neighborhood

We observe that the pixel-level AC score in $\mathbf{C}$ is sensitive to the local noises, e.g., in Fig. 5(i), a normal image that contains several small noises has a higher AC score $c = 0.129$, while the pure normal image (ii) has a lower score $c = 0.079$, when we calculate the class token distance between images (i) and (ii), they are almost the same with a similarity equal to 0.987, which reveals that the class token distance is less sensitive to local noises, and two images that have a higher class token similarity should have an approximate AC score. A similar situation can be observed in an abnormal image (iv), where the image contains a low contrast abnormal area and has a lower AC score in (iv), while it has a higher similar to the abnormal image (iii).

Motivated by such an observation, we propose the re-scoring with constrained image-level neighborhood to optimize the anomaly classification by exploiting the relationship between the image-level features of images in $\mathcal{D}_u$. Specifically, given the class token $F_i \in \mathbb{R}^{1 \times C}$ of the last layer of ViT, we

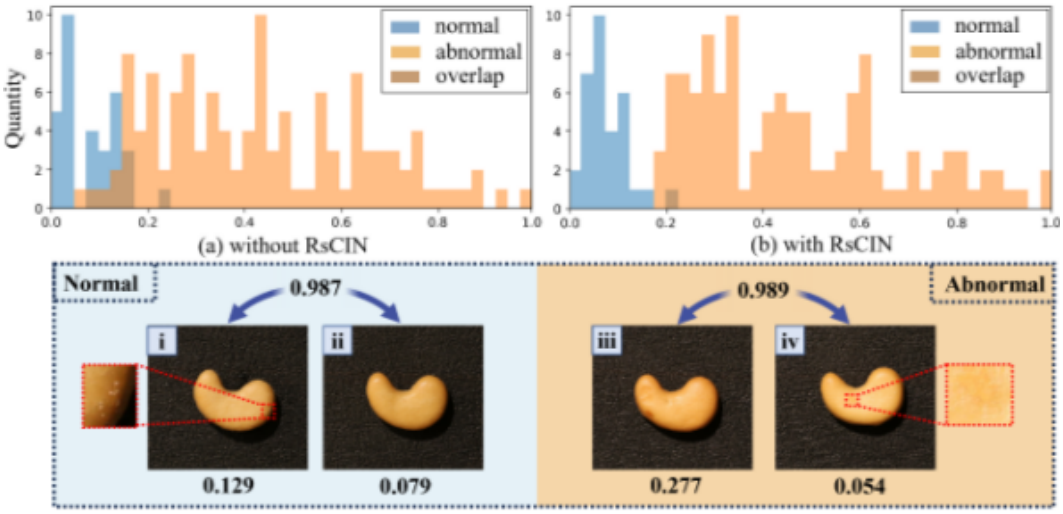


Figure 4: **Top:** Histogram of anomaly classification scores of unlabeled test images before (a) and after (b) using RsCIN, with blue representing normal images and orange representing abnormal images. **Bottom:** A normal and an abnormal example of RsCIN.

define an edge-weighted graph $G = (V, W)$, the vertexes V denotes the images in $\mathcal{D}_u$, and the edge weight W is defined as the similarity of the class tokens F_i of I_i, i.e., $W_{i,j} = F_i \cdot F_j$, where $\cdot$ means dot product operation. Therefore, the manifold learning approaches, e.g., SSO (Jiang et al., 2011), can be employed for optimizing $\mathbf{C}$ by utilizing the token similarity W.

However, due to the high smoothness of $\mathbf{C}$, it may be influenced by a large number of images, as the experimental illustration in Sec. 4.2, excessive utilization of the images in $\mathcal{D}_u$ may reduce the AC accuracy. Therefore, we propose a Multi-window Mask Operation (MMO) to constrain the image number, which makes each image only influenced by a small number of neighborhood images. To be specific, we design a binary window mask matrix $M_k \in \mathbb{R}^{N \times N}$ as,

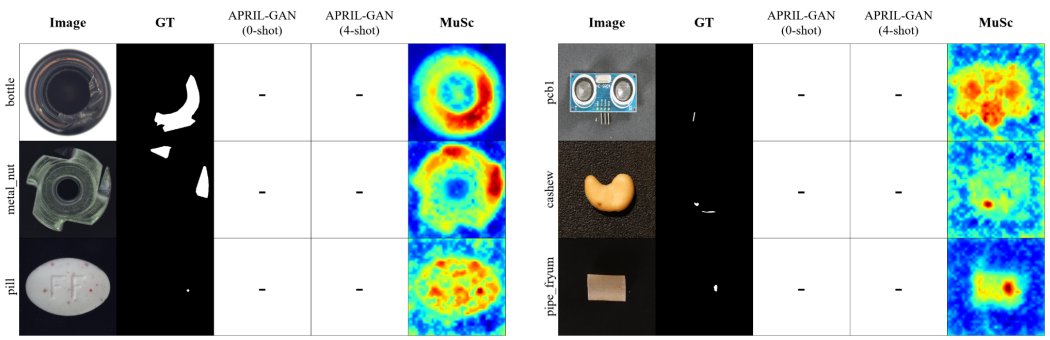
$$M_k(i,j) = \begin{cases} 1, & \text{if } I_j \in \mathcal{N}_k(I_i), \\ 0, & \text{otherwise}, \end{cases} \tag{4}$$

$\mathcal{N}_k$ indicates the k-nearest neighbor. We use multiple k to form the multi-window mask set $\bar{\mathcal{M}} = \{M_{k_1}, ..., M_{k_K}\}$, where K is the number of window masks. Hence the AC score $\mathbf{C}$ is updated as,

$$\hat{\mathbf{C}} = \left(\sum_{M_k \in \bar{\mathcal{M}}} \left(D^{-1}(M_k \odot W)\mathbf{C} \right) + \mathbf{C} \right) / (K+1) \tag{5}$$

where $\hat{\mathbf{C}} \in \mathbb{R}^{N \times 1}$ is the optimized AC score vector, D is a diagonal matrix with $D(i,i) = \sum_{j=1}^{N} M_k \odot W(i,j)$ and $\odot$ means element-wise multiplication. Eq. 5 utilizes the weighted AC scores of multiple neighbor images of I_i to refine $\mathbf{C}$. Therefore I_i is normal only if itself and its k nearest neighbor images all have a small AC score. As shown in Fig. 5, the AC scores of normal and abnormal images in $\mathbf{C}$ have a large overlap in (a), while in (b), the overlap is reduced in $\hat{\mathbf{C}}$. See Appendix A.1.2 for more details. Meanwhile, we experimentally prove that the proposed image-level AC optimization method can be further employed to improve the AC accuracy of existing approaches in Appendix A.2.4.

Dataset	Method	Setting	AUROC-cls	F1-max-cls	AP-cls	AUROC-segm	F1-max-segm	AP-segm	PRO-segm
MVTec AD	WinCLIP (Jeong et al., 2023)	0-shot	90.4	92.7	95.6	82.3	24.8	18.2	61.9
	APRIL-GAN (Chen et al., 2023)	0-shot	86.1	90.4	93.6	87.6	43.3	40.8	44.0
	ACR (Li et al., 2023)	0-shot	86.6	91.9	93.6	92.6	44.3	38.6	74.0
	MuSc (ours)	0-shot	**97.8**	**97.3**	**99.1**	**97.1**	**62.2**	**62.3**	**93.5**
VisA	WinCLIP (Jeong et al., 2023)	0-shot	75.5	78.2	78.7	73.2	9.0	5.4	51.0
	APRIL-GAN (Chen et al., 2023)	0-shot	77.5	78.6	80.9	94.2	32.3	25.8	86.6
	MuSc (ours)	0-shot	**92.5**	**88.9**	**93.3**	**98.7**	-	-	-
MVTec AD	RegAD (Huang et al., 2022)	4-shot	88.8±1.0	92.2±0.3	94.7±0.8	96.1±0.2	51.3±0.8	47.9±0.9	87.9±0.4
	PatchCore (Roth et al., 2022)	4-shot	89.9±0.4	93.9±0.2	95.2±0.2	94.3±0.4	53.2±0.7	52.3±0.6	83.0±0.6
	WinCLIP (Jeong et al., 2023)	4-shot	95.3	94.9	97.6	94.2	45.9	42.2	85.4
	APRIL-GAN (Chen et al., 2023)	4-shot	92.7±0.1	92.8±0.1	96.2±0.0	95.9±0.0	56.7±0.5	54.4±0.3	91.9±0.0
	GraphCore (Xie et al., 2023)	4-shot	81.0±1.1	89.4±0.3	89.8±0.6	88.8±0.5	36.6±0.6	27.7±0.6	75.4±0.5
	MuSc (ours)	0-shot	**97.8**	**97.3**	**99.1**	**97.1**	**62.2**	**62.3**	**93.5**
VisA	PatchCore (Roth et al., 2022)	4-shot	83.1±1.3	83.4±1.0	85.3±0.7	96.1±0.1	39.3±1.0	35.2±1.4	76.9±0.3
	WinCLIP (Jeong et al., 2023)	4-shot	84.1	82.4	85.8	95.2	25.3	17.7	81.4
	APRIL-GAN (Chen et al., 2023)	4-shot	92.6±0.1	88.2±0.3	94.5±0.1	96.2±0.0	39.6±0.3	32.0±0.2	90.1±0.1
	MuSc (ours)	0-shot	**92.5**	**88.9**	**93.3**	**98.7**	-	-	-



(a) MVTec AD (b) VisA

Figure 6. Local comparison and qualitative examples.

4 EXPERIMENTS

Implementation Details. In this paper, we use ViT-L/14-336 pre-trained by OpenAI (Radford et al., 2021) as our backbone. It contains 24 layers, which are divided into 4 stages. Each stage is composed of 6 layers. We extract the patch tokens output from each stage for our method and use the linearly projected class token of the last layer for classification optimization. All the unlabeled test images are scaled to a resolution of 518×518 and then fed into the backbone. In the MSM module, we use the minimum 30% anomaly scores. In the RsCIN module, we use the multi-window setting with 1, 2, 3 on MVTec AD and 8, 9 on VisA.

Datasets. To study Industrial anomaly classification and segmentation performance, we conduct experiments on industrial image datasets MVTec AD (Bergmann et al., 2019) and VisA (Zou et al., 2022). MVTec AD has high-resolution (from 700^2 to 1024^2) RGB images of 10 object categories and 5 texture categories. VisA comprises high-resolution (1000×1500) RGB images covering 12 objects in 3 domains. Both benchmarks contain normal and anomaly images in the test images.

Evaluation Metrics. For classification metrics, we report 3 metrics: the Area Under Receiver Operator Characteristic curve (AUROC), Average Precision (AP), and F1-score at optimal threshold (F1-max). For segmentation metrics, we report 4 metrics: pixel-wise AUROC, pixel-wise F1-max, pixel-wise AP, and Per-Region Overlap (PRO).

Baselines. We compare our method with some state-of-the-art zero/few-shot approaches, e.g. WinCLIP (Jeong et al., 2023), APRIL-GAN (Chen et al., 2023), RegAD (Huang et al., 2022), PatchCore (Roth et al., 2022), GraphCore (Xie et al., 2023) and ACR Li et al. (2023). Among these methods. APRIL-GAN is in 1st place for the Zero-shot Track of the CVPR 2023 VAND Workshop Challenge, which uses an additional labeled auxiliary dataset for training. RegAD and ACR use other categories of the dataset when training. The unmeasured metrics in these papers are reproduced by using the official implementation code. In addition, we also compare our method with some full-shot methods.

Table 2: Comparison with some existing many-shot methods in image-AUROC and pixel-AUROC (%) on MVTec AD.

Methods	Setting	AC	AS
CutPaste (Li et al., 2021)	full-shot	95.2	88.3
NSA (Schlüter et al., 2022)	full-shot	97.2	96.3
IGD (Chen et al., 2022)	full-shot	93.4	93.0
PatchCore (Roth et al., 2022)	full-shot	99.6	98.2
RegAD (Huang et al., 2022)	32-shot	96.4	96.9
GraphCore (Xie et al., 2023)	8-shot	95.9	97.8
MuSc(ours)	0-shot	97.8	97.3

Table 3: Ablation study of LNAMD with different aggregation degrees r in image-AUROC and pixel-AUROC (%) on MVTec AD and VisA.

	MVTec AD		VisA	
r	AC	AS	AC	AS
$\{1\}$	96.8	94.6	93.0	97.9
$\{3\}$	97.3	97.1	89.5	98.4
$\{5\}$	95.4	97.6	82.2	97.6
$\{1,3\}$	98.1	96.3	93.7	98.6
$\{3,5\}$	96.5	97.8	87.2	98.3
$\{1,3,5\}$	97.8	97.3	92.8	98.7

Table 4: Ablation study of MSM with different sample strategies in image-AUROC and pixel-AUROC (%) on MVTec AD and VisA.

	MVTec AD		VisA	
setting	AC	AS	AC	AS
(a)min	96.8	94.1	91.0	98.4
(b)max	83.8	92.9	74.9	95.2
(c)mean	95.0	96.8	89.6	98.4
(d)30% + min	96.8	94.1	91.0	98.4
(e)30% + max	96.0	**97.5**	91.2	98.6
(f)30% + mean	**97.8**	97.3	**92.8**	**98.7**

Table 5: Ablation study of RsCIN. We report anomaly classification performance in AUROC, F1-max and AP on MVTec AD and VisA dataset. All metrics are in %. Bold indicates the best performance.

Datasets	RsCIN	AUROC	F1-max	AP
MVTec AD	w/o	97.4	96.6	98.7
	w	**97.8**	**97.5**	**99.1**
VisA	w/o	90	87.1	90.9
	w	**92.8**	**89.5**	**93.5**

4.1 Quantitative and Qualitative Results

Comparison with zero/few-shot methods. In Tab. 1, we compare MuSc with some state-of-the-art zero-shot methods on MVTec AD and VisA datasets. On MVTec AD, MuSc achieves 19.5% improvement on PRO and 21.9% improvement on AP than the second-best method in the zero-shot setting. On VisA, in classification and segmentation, our method has more than 10% improvement compared with the second-best method. By comparing with some 4-shot methods, the classification and segmentation results of MuSc beyond most methods with four normal reference images, although we do not use any labeled images.

We show the industrial image visualization of anomaly maps of MuSc in Fig. 6. Compared with other zero/few-shot methods, our method can detect small defects and has fewer false positives, as shown in the pill and cashew. Other methods in zero/few-shot are sensitive to the normal spots on the pill and the tiny splinters on the cashew. To the large defects shown in bottle and pcb1, our method can segment the whole regions. Some logical anomalies, such as the flip of metalnut and the proliferation of pipe fryum, can be segmented entirely by our method.

Comparison with many-shot methods. We compare our MuSc with some many-shot methods on MVTec AD, shown in Tab. 2. MuSc is comparable to some full-shot methods, such as CutPaste (Li et al., 2021), IGD (Chen et al., 2022) and NSA (Schlüter et al., 2022), which utilize all the normal reference images in the train set. Without using any labeled images, our method also outperforms RegAD with 32-shot and is competitive with GraphCore with 8-shot.

4.2 Ablation Study

The influence of the aggregation degree r. The aggregation degree r controls the size of the local neighborhood to be aggregated in the LNAMD module. As reported in Tab. 3, we conduct the experiments with combinations of different aggregation degrees. The experimental results

Table 6: Per-image inference time and maximum GPU memory cost by our MuSc. We divide the whole dataset into s subsets, measured at NVIDIA GeForce RTX 5080 GPU. We report the time and GPU memory cost with different s.

s	Time (ms/image)	GPU cost (MB)
1	726.5	7418.2
2	420.1	7070.7
3	327.7	6804.2

Table 7: The effect of dataset size. We divide the dataset into s parts, evaluate image-AUROC and pixel-AUROC(%) on MVTec AD and VisA.

Datasets	s	AC	AS	Size of the subsets
MVTec AD	1	97.8	97.1	42–167
	2	97.6	97.1	21–86
	3	97.4	97.1	14–58
VisA	1	92.5	98.7	150–201
	2	92.5	98.7	75–101
	3	92.4	98.7	49–68

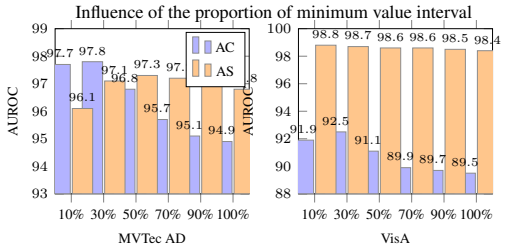


Figure 7: The influence of the proportion of minimum value interval selection in MSM.

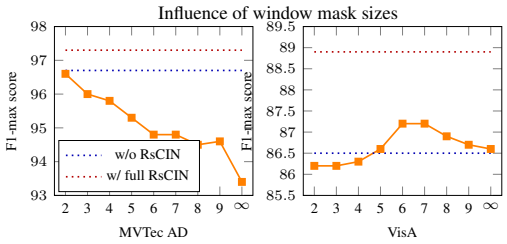


Figure 8: The influence of different window mask sizes on MVTec AD and VisA.

demonstrate that using all three aggregation degrees of $r \in \{1, 3, 5\}$ is better than other cases in the comprehensive results of AC and AS. Moreover, due to the different datasets with varying anomaly sizes, a smaller r is suitable for segmenting small abnormal regions, which is common on VisA, while a larger r does well in segmenting large abnormal regions, which is common on MVTec AD. To balance the representation on different datasets, we use three scales of aggregation degrees.

Discussion of the mutual scoring mechanism. Here, we modify Eq. 2 in six different cases, shown in Tab. 4. In (a), (b), and (c), we remove the operation of selecting the minimum 30% value interval. (a) and (d) means replace the average operation with the minimum operation, and in (b) and (e) we replace it with the maximum operation. (f) is the optimal setting for our method. We also test the effect of the selection percentage shown in Fig. 7. Selecting the minimum 30% can bring better comprehensive AC and AS results on the two datasets.

Discussion of the re-scoring with constrained image-level neighborhood. In Tab. 5, we remove the re-scoring with constrained image-level neighborhood module to check our method for classification. The results demonstrate the RsCIN module could bring 0.9% F1-max-score gains on MVTec AD and 2.8% AUROC gains on VisA. Meanwhile, for the multi-window mask operation of RsCIN, we replace the multi-window with the single-window $k \in \{2, \ldots, 9\}$ and full-window $k = \infty$ without the mask operation in Fig. 8. Our multi-window strategy outperforms the single-scale strategy on both datasets. In addition, we insert our RsCIN module into some other different AC/AS methods, and the classification results have been improved, detailed in Appendix A.2.4.

Discussion of inference time and memory cost. The inference time and memory cost are also what we are concerned about. In our MuSc, we report the per-image inference time and maximum GPU memory cost in Tab. 6, where the memory cost is measured on the most numerous dataset which contains 200 images. When the number of test images increases, the inference time and memory cost continue to increase, which may become a limitation of our method.

To break through this limitation, we divide the entire test images into s subsets, calculate the anomaly scores on every subset, and finally compute the AC and AS metrics together. The AC/AS results are shown in Tab. 7, where 42–167 means the maximum size of the divided subset is 167 and the minimum size is 42. As the subset size decreases, the performance decreases by less than 1.1% for AC and 0.2% for AS, which means our method works well even with tiny datasets. The inference time and memory cost also decrease as the subset size decreases, shown in Tab. 6. Therefore, when

the test set is large, we consider using our MuSc in its subsets separately. Additionally, we conduct a comprehensive efficiency comparison in Appendix A.4.

Discussion of few-shot setting. Our MuSc is a pure zero-shot method that does not require any labeled images. We design two extension methods in the few-shot setting based on our method when a small number of normal reference images are provided, see Appendix A.6. There is no significant improvement in the results when a few images are added because our method already uses a large amount of normal prior information in the unlabeled test images.

5 Conclusion

In this paper, we propose a novel zero-shot industrial AC and AS framework named MuSc. We explore the normal and abnormal cues implicit in the unlabeled test images. First, we fuse local neighborhood aggregated patch tokens in different degrees. Then, we use a mutual scoring mechanism for each image. Finally, we propose an image-level AC optimization method. Notably, our method surpasses the existing zero-shot approaches, even outperforms most of the few-shot approaches, and is comparable to some full-shot methods.

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